

Karl Rodriguez

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TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, HTML, CSS, SQL, R

Frameworks & Tools: Node.js, React, Spring Boot, Jest, JUnit, Selenium, MS SQL Server, PostgreSQL

EDUCATION

Simon Fraser University | Bachelor of Science - Computer Science

September 2023 - April 2027

PROJECTS

Business Review & Social Database System | Python, MS SQL Server **April 2026**

- Built a Python application connected to a Microsoft SQL Server Yelp-style database using pyodbc
- Implemented SQL queries for searching businesses and users with filters
- Enabled users to post reviews and manage friendships through database operations and transactions
- Designed and utilized SQL triggers to automate updates and maintain data consistency

AppTrack | TypeScript, Java, PostgreSQL, React, Spring Boot, Jest, JUnit **December 2025**

- Developed a full-stack application to track submitted applications, building both frontend and backend features
- Created RESTful APIs in Java with Spring Boot and connected PostgreSQL for data storage
- Wrote unit and integration tests using JUnit and Jest to verify API endpoints and frontend functionality
- Found and fixed defects to maintain reliable application performance

Study Buddy | React, JavaScript, HTML, CSS, Jest **September 2025**

- Managed quality assurance for an AI-powered study assistant using the Google Gemini API
- Wrote and maintained Jest unit tests for frontend components and API interactions
- Set up a CI/CD pipeline to automate testing and deployment
- Tracked and reported bugs using Jira and worked with the team to resolve issues

Emergency Report | React, Bootstrap, TypeScript, HTML, CSS, Jest **June 2025**

- Led development of a real-time emergency reporting web application with interactive mapping
- Tested core functionality with Jest and verified form validation and input handling
- Created test cases to ensure components behaved correctly

Event Calendar | Java, Spring Boot, JUnit **January 2025**

- Developed a RESTful API for creating and viewing scheduled events
- Used JUnit to test backend logic and API endpoints

Find Recipes | React, TypeScript, HTML, CSS, Jest, Selenium **November 2024**

- Built a recipe recommendation application to help users discover new recipes
- Wrote unit tests with Jest to ensure workflows and data handling were accurate
- Developed automated end-to-end tests using Selenium to validate core user workflows